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MILITARY SPECIFICATION

# INDICATOR, RATE OF FLOW, FUEL, TOTALIZING, 0-100,000 PPH, TYPE MA-8

## 1. SCOPE

1.1 This specification covers a remote-indicating summation fuel-flow indicator designated Type MA-8.

## 2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on the date of invitation for bids, form a part of this specification to the extent specified herein:

SPECIFICATIONS

Military

| | |
|---|---|
| MIL-I-7566 | Indicator System, Fuel Flowmeter, General Specification for |
| MIL-I-25324 | Indicator, Rate of Flow, Fuel, Totalizing, 0-12,000 PPH, Type MA-7 |

STANDARDS

Military

| | |
|---|---|
| MIL-STD-130 | Identification Marking of U.S. Military Property |
| MS33585 | Pointers, Dial, Standard Design of Aircraft Instrument |
| MS33639 | Cases, Instrument, Clamp-Mounted, Aircraft Instrument |
| MS33678 | Connectors, Receptacle, Electric, Integral Mounting |

(Copies of specifications, standards, drawings and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

FSC 6620

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## 3. REQUIREMENTS

3.1 QUALIFICATION.- The indicator furnished under this specification shall be a product which has been tested, and passed the qualification tests herein, and has been listed on or approved for listing on the applicable qualified products list.

3.2 GENERAL SPECIFICATION.- The requirements of MIL-I-7566 apply as requirements of this specification with the exceptions and additions called out herein. When two specifications conflict, this specification shall govern.

3.3 DESIGN AND CONSTRUCTION

3.3.1 The summation flow-rate indicator shall be designed with dual pointers to indicate a rate-of-fuel flow from 0 to 100,000 pph. The fast pointer shall indicate 10,000 pph per revolution and the slow pointer 100,000 pph in one revolution.

3.3.1.1 The indicator shall operate from an interrupted 400-cycle voltage of 1V per 1000 pph flow rate supplied by 2 or more Type MA-7 rate-of-flow indicators in accordance with MIL-I-25324.

3.3.1.2 The indicator section of the indicator shall contain a low-inertia 2-phase motor, other components, and pointers to indicate totalized rate of flow.

3.3.1.3 An amplifier shall be incorporated as an integral unit contained within the indicator case.

3.3.1.4 The indicator shall be designed to indicate a change of flow rate from 0 to 100,000 pph in not more than 15 seconds at room temperature.

3.4 CASE.- The case shall conform to figure 1 and shall be sealed hermetically.

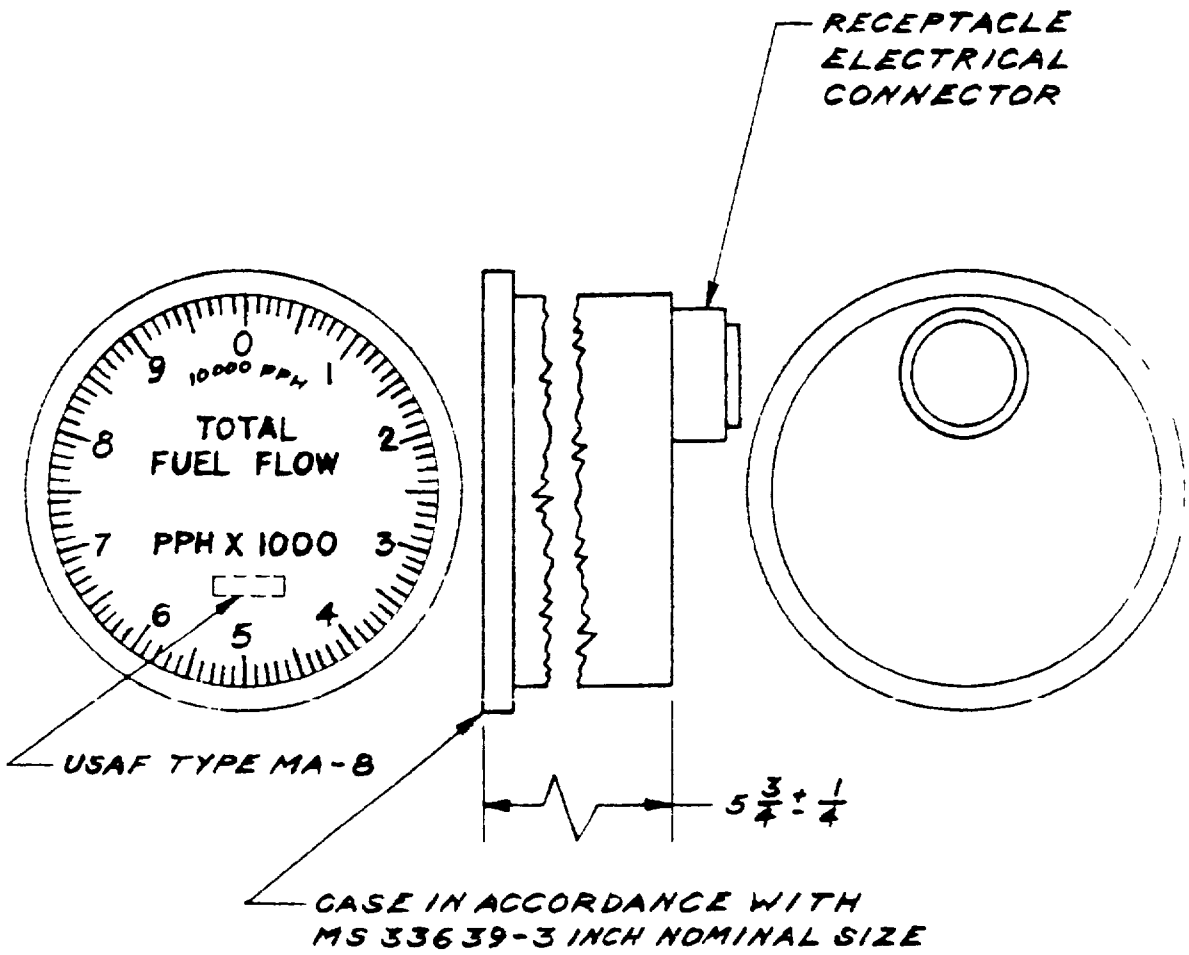
3.5 DIAL

3.5.1 The minimum diameter of the dial, as measured across the outside ends of the graduations, shall be 2-29/32 inches. The dial for the summation flow-rate indicator shall conform to figure 1.

3.5.2 DIAL-TO-COVER-GLASS DISTANCE.- The distance between the indicator dial and the inside of the cover glass shall be the minimum practicable and shall not exceed 0.187 inch.



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DIMENSIONS IN INCHES

FIGURE 1 DIAL AND CASE



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3.6 POINTERS.- The short pointer shall conform to MS33585-4 and the long pointer to MS33585-3, except that the length shall be such that the tip will overlap the shortest graduation by 1/5 of the graduation length. The taper of the tip of the long pointer shall be 5/16 inch.

3.7 POWER INPUT.- The power supply shall be 115 ±5V, 400 ±20 cycles, single phase. The waveform of the supply voltage may be nonsinusoidal with a crest factor of 1.414 ±10 percent and with harmonics whose respective amplitudes are less than 5 percent of the fundamental. No damage shall result to the instrument if a voltage or frequency of 10 percent exceeding the maximum or minimum tolerance specified is applied.

3.8 CABLE LENGTH.- Accuracy and sensitivity of the indicator system shall not be adversely affected by varying the length of the interconnecting cable between the rate-of-flow indicators and the total rate-of-flow indicator for a cable capacitance up to 1200 uuf.

3.9 CRUISE CONTROL OUTPUT.- The indicator shall contain a potentiometer whose slider is positioned proportional to indication. The potentiometer terminals shall be brought directly to the connector with no other internal connections. External connections to this potentiometer shall cause no more than 1 watt of power to be dissipated inside the instrument.

3.10 WEIGHT.- The weight of the indicator and self-contained amplifier shall not exceed 2.5 pounds.

3.11 ELECTRICAL RECEPTACLES.- The indicator shall be provided with an electrical receptacle in accordance with MS33678-18-1PW, except that it shall be designed for hermetic sealing.

3.12 MARKINGS

3.12.1 FLUORESCENT-LUMINESCENT.- The following markings shall be finished in fluorescent-luminescent material. The dimensions of the markings shall be as follows:

| | Height or Length Inch ±1/64 | Width or Line or Graduation Inch ±.005 |
|---|---|---|
| Scale numerals 1, 2, 3, etc. | 7/32 | .031 |
| Graduations opposite numerals | 7/32 | .031 |
| 500-pound graduations | 5/32 | .020 |
| 100-pound graduations | 3/32 | .016 |
| Lettering "PPH x 1000" | 1/8 | .020 |
| Lettering "10,000 PPH" | 1/16 | .016 |
| Lettering "TOTAL FUEL FLOW" | 1/8 | .020 |



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3.12.2 DURABLE DULL BLACK.- The following markings shall be marked permanently and legibly on the dial. These markings and all other markings not otherwise specified shall be furnished in durable dull black. The height of the markings shall be as follows:

| | Height Inch |
|---|---|
| Lettering "USAF" and "TYPE MA-8" | 3/64 |

3.13 IDENTIFICATION OF PRODUCT.- Equipment, assemblies, and parts shall be marked for identification in accordance with MIL-STD-130.

4. QUALITY ASSURANCE PROVISIONS

4.1 SUPPLIER RESPONSIBILITY.- The supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own or any other inspection facilities and services acceptable to the Government. Inspection records of the examination and tests shall be kept complete and available to the Government as specified in the contract or order. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 GENERAL SPECIFICATION.- The sampling, inspection, and test procedures shall be in accordance with MIL-I-7566 and as specified herein.

4.3 TEST EQUIPMENT STANDARDS

4.3.1 MASTER TRANSMITTER.- Indicators shall be tested with a master transmitter as specified in MIL-I-7566 except that for system indication, the master transmitter shall provide an output equivalent to the summation output signal of 1 to 8 individual rate-of-flow indicators.

4.4 PERFORMANCE TESTS.- The performance of the indicator shall be checked against tables I, II, and III of this specification and shall not exceed the tolerances specified therein.



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TABLE I

| INDICATOR SCALE ERROR | | | | |
|---|---|---|---|---|
| Flow Rate PPH | Room Temp Tol ± PPH | Low Temp Tol +PPH | Low Temp Tol -PPH | High Temp Tol ±PPH |
| 2,000 | 100 | 100 | 225 | 125 |
| 5,000 | 100 | 100 | 240 | 125 |
| 10,000 | 100 | 100 | 280 | 125 |
| 25,000 | 100 | 125 | 300 | 185 |
| 30,000 | 150 | 150 | 320 | 225 |
| 40,000 | 200 | 200 | 380 | 350 |
| 50,000 | 240 | 240 | 440 | 450 |
| 60,000 | 280 | 280 | 500 | 550 |
| 70,000 | 320 | 320 | 560 | 650 |
| 80,000 | 360 | 360 | 620 | 750 |
| 90,000 | 400 | 400 | 680 | 850 |
| 100,000 | 440 | 440 | 740 | 950 |

TABLE II

| INDICATOR FRICTION AND VIBRATION ERROR | |
|---|---|
| Test | Tolerance ±PPH |
| Position error* | 100 |
| Vibration | |
| Pointer oscillation-double amplitude | 50 |
| Pointer variation | 50 |
| Vibration failure | |
| Scale error | Room temp limits of table I |
| Friction error | 50 |
| Friction | |
| Room temperature | 50 |
| High temperature | 50 |
| Low temperature | 100 |
| Response time | 15 sec max |
| Power input | 20 watts max |

*Position error at 20,000 and at 70,000 PPH



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TABLE III

| VOLTAGE AND FREQUENCY VARIATION TEST | |
|---|---|
| Flow Rates | Tolerance ±PPH |
| 0-25,000 | 150 |
| 30,000 | 200 |
| 40,000-100,000 | 300 |

4.5 SAMPLING PLAN B TESTS

4.5.1 ENDURANCE.- The indicator shall be subjected to the endurance test of MIL-I-7566 except that the number of cycles per minute shall be 25,000 cycles in 1200 hours. The scale error after test shall not exceed the low and high temperature limits of table I.

4.5.2 SENSITIVITY.- The indicator shall be connected to the master transmitter in a manner to provide an indication. A change shall be made in the transmitter setting equivalent to 500 pph. A change of 500 ±100 pph shall be shown on the indicator in the range from 0-30,000 and a change of 500 ±150 pph shall be shown in the range from 30,000 to 100,000. This test may be conducted at any indication.

4.5.3 CRUISE-CONTROL OUTPUT ERROR.- The resistance of cruise control output shall be in the same proportion of the total potentiometer resistance as the scale reading is of full scale, within the following limits. For a full scale value of 100,000 pph, the deviation from scale error shall be less than ±100 ppn in the range from 20,000 to 40,000 pph and ±250 pph above and below this range.

4.5.4 INDIVIDUAL TESTS

4.5.4.1 SYSTEM TEST.- A room-temperature system test shall be conducted at indicated flow rates, utilizing summation output signals of eight individual indicators. A master transmitter, per MIL-I-7566, or other suitable means, shall be used to actuate the individual indicators. The indicated summation flow rate shall equal the sum of the indicated individual flow rates within ±2 percent tolerance at the 16,000, 24,000, 48,000 and 72,000 pph test points.

5. PREPARATION FOR DELIVERY

5.1 PREPARATION.- The indicator shall be packaged and packed for shipment in accordance with MIL-I-7566.



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5.2 MARKING OF SHIPMENTS.- Marking of shipments shall be in accordance with MIL-I-7566. The nomenclature shall be as follows:

Indicator, Rate of Flow, Fuel, Totalizing, 0-100,000 PPH, 115V, 400 Cycle, AC, Type MA-8

5.2.1 REINSPECTION MARKINGS.- The reinspection date markings shall be as specified by the procuring activity.

6. NOTES

6.1 INTENDED USE.- The Type MA-8 indicator covered by this specification is intended for use to indicate the total fuel flow rate indication for multiengine aircraft.

6.2 ORDERING DATA.- Procurement documents should specify the following:

a. Title, number, and date of this specification.

b. Whether sampling plan B tests are to be conducted.

c. Number of qualification samples and point of testing.

d. Level of packaging and packing required.

6.3 PROVISIONS FOR QUALIFICATION.- With respect to products requiring qualification, awards will be made only for such products as have, prior to the time set for opening of bids, been tested and approved for inclusion in the applicable qualified products list whether or not such products have actually been so listed by that date. The attention of suppliers is called to this requirement, and manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government tested for qualification in order that they may be eligible to be awarded contracts or orders for the products covered by this specification. The activity responsible for the qualified products list is Aeronautical Systems Division and information pertaining to qualification of products covered by this specification may be obtained from Aeronautical Systems Division (ASNPF), Wright-Patterson Air Force Base, Ohio.



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